ResQAI - Disaster Management System

System Audit, Test Verification & Deployment Readiness Report

1. Executive Summary & Verification Status

The AI Disaster Reporting & Triage system has been comprehensively tested and hardened. The system now strictly discriminates between authentic emergencies and fake reports (pranks, jokes, metaphors, fictional claims, spam, and vague abstractions). All 8 automated test suites have passed with zero regressions.

Test Module	Result	Validation Scope
Duplicate Detection	**PASSED [100%]**	Spatial (30m) & temporal (15m) deduplication active
Prank & Fake Rejection	**PASSED [100%]**	7/7 fake & prank scenarios caught and rejected with HTTP 400
Genuine Disaster Intake	**PASSED [100%]**	12/12 genuine disasters correctly classified & ingested
Image Verification	**PASSED [100%]**	Corrupt, blank, and micro-resolution photos rejected
AI Fallback & Retries	**PASSED [100%]**	Crash-proof retry loop and deterministic offline fallback
Low-Bandwidth & WAL	**PASSED [100%]**	Gzip compression and SQLite WAL concurrency verified
Offline Batch Sync	**PASSED [100%]**	Multi-report queue ingest with deduplication pipeline
Frontend Production Build	**PASSED [100%]**	Vite production build succeeds in 1.16s with 0 errors

2. Architectural Notes & Production Considerations

1. **Image Storage (Local Disk /uploads):** Incident photos are saved locally. For persistent multi-instance serverless deployments (e.g. Render Free Tier), persistent block storage or an S3/Cloudinary bucket should be attached.
2. **Database Layer (SQLite WAL):** Current engine uses SQLite in Write-Ahead-Logging mode (zero setup, ideal for hackathon evaluation and demonstrations). For thousands of concurrent write transactions, PostgreSQL can be configured.
3. **GIS Map Tiles:** Leaflet / OpenStreetMap tiles fetch dynamically via web. Report ingestion and offline batch sync remain fully operational even if tile fetching is constrained.
4. **Nugen AI Timeout Window:** A 120-second timeout is configured to accommodate high-latency LLM runs. Offline heuristic algorithms immediately resolve triage scores if the LLM connection fails.

3. Step-by-Step Execution Guide

Terminal 1 - Backend Server:

```
uvicorn main:app --reload --port 8000
```

Terminal 2 - Frontend Client:

```
cd frontend && npm run dev
```

Terminal 3 - Run Automated Test Suite:

```
python run_all_tests.py
```

4. Built-in Authority Credentials for Demo

Role	Username	Password	Access Level
Command Chief	responder.chief	emergency2026	Full Dispatch & Status Modification
Field Officer	demo	demo123	Standard Responder View